

HORNSBY GIRLS' HIGH SCHOOL



MATHEMATICS

YEAR 11

Preliminary Assessment Task 2

Half-Yearly Examination 2010

Student Number: _____

Time Allowed: 90 minutes plus 5 minutes reading time

Instructions:

- Attempt all questions
- Start a new page for each question
- The marks for each question are indicated
- Show all necessary working
- Marks may be deducted for untidy or badly arranged work
- Board approved calculators may be used

Outcomes Assessed:

- P1 Demonstrates confidence in obtaining realistic solutions to problems
- P2 Provides reasoning to support conclusions in the correct context
- P3 Performs routine arithmetic and algebraic manipulation involving surds, simple rational expressions and trigonometric identities
- P4 Chooses and applies appropriate arithmetic, algebraic, graphical, trigonometric and geometric techniques
- P5 Understands the concept of a function and the relationship between a function and its graph

Outcomes	Total	
Algebra and Arithmetic	Q1.	/13
Functions and Graphs	Q2.	/13
	Q3.	/10
Linear Functions	Q4.	/12
Trigonometric Functions	Q5.	/12
	Q6.	/12
Total Marks:		/72

This assessment task constitutes 30% of the final Preliminary Course Assessment.

(A) Evaluate correct to 2 decimal places:

2

$$\frac{1}{(2.4)^2} + \frac{1}{\sqrt{3.7}}$$

(B) Rationalise the denominator of $\frac{1 + \sqrt{xy}}{x\sqrt{x}}$

2

where x and $y \geq 0$

(C) Factorise completely: $2p^3 + 54$

2

(D) Expand $(h - 2)^3$

2

(E) Find all values of x which satisfy

2

$$|3x - 4| + 2 \leq 7$$

(F) Find rational numbers a and b such that

3

$$a + b\sqrt{3} = \frac{\sqrt{3}}{3 + \sqrt{3}}$$

(A) State the domain for the following

(i) $y = \frac{x^2}{x^2 - 4}$ 2

(ii) $f(x) = \frac{1}{\sqrt{3-x}}$ 1

(B) If $g(x) = \frac{x^2}{x^4 + 1}$

(i) find $g(-x)$ 1

(ii) state why $g(x)$ is an even function. 1

(C) If $h(x) = x^2 + 3x$ find:

(i) $h(-2)$ 1

(ii) $h(a)$ 1

(iii) $h(x-1)$ 2

(D) By sketching $q(x) = x(x-1)^2$ or otherwise, solve $q(x) \geq 0$. 2

(E) Sketch the function $f(x) = |x| + x$ for all x . 2

QUESTION 3**(START NEW PAGE)****10 MARKS**

(A) On separate diagrams sketch the following curves

(i) $y = x^2 + 2x - 3$ 2

(ii) $y = \sqrt{4 - x^2}$ 2

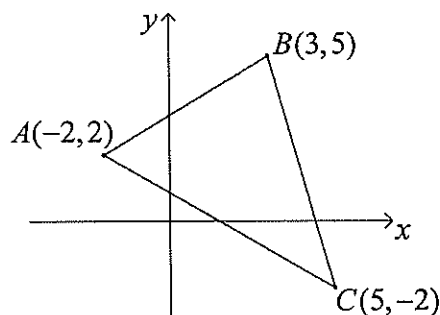
(B) Sketch the region defined by $y > x - 2$ and $y \leq 4 - x^2$ 3

(C) Sketch the graph of $f(x)$ if 3

$$f(x) = \begin{cases} 2x-1 & \text{for } x \geq 0 \\ x^2 & \text{for } x < 0 \end{cases}$$

QUESTION 4**(START NEW PAGE)****12 MARKS**

(A) Consider the following $\triangle ABC$



(i) Find the exact distance AC . 1

(ii) Show that the equation of the line through A and C is $4x + 7y - 6 = 0$ 2

(iii) Find the altitude of $\triangle ABC$ through vertex B . 2

(iv) Hence or otherwise, find the area of $\triangle ABC$. 1

(B) Find the equation of the straight line passing through the point $(1, 2)$ and the point of intersection of the lines $3x - 5y - 3 = 0$ and $2x + 3y + 17 = 0$. 3

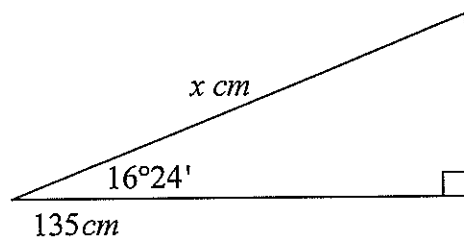
(C) If $2x - 3y - 3 = 0$ and $3x + ay + b = 0$ are perpendicular and intersect at $x = 3$, find the values of a and b . 3

(A) Find θ to the nearest minute if θ is an acute angle and

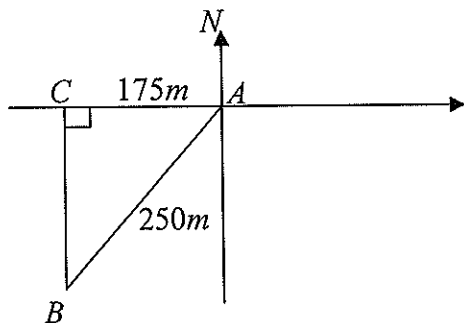
(i) $\sin \theta = \frac{3}{4}$ 1

(ii) $\sec \theta = 1.2230$ 1

(B) Find x correct to one decimal place. 2



(C) Find the bearing of B from A to the nearest degree. 2



(D) Find the exact value of

(i) $\tan 210^\circ$ 1

(ii) $\sin 330^\circ$ 1

(iii) $\sin 45^\circ \cos 120^\circ$ 2

(E) If $\sec \alpha = \frac{7}{3}$ and $\sin \alpha < 0$ find the exact value of $\tan \alpha$. 2

QUESTION 6**(START NEW PAGE)****12 MARKS**

(A) Draw sketches of the following for $0^\circ \leq \theta \leq 360^\circ$

(i) $y = \sin \theta$ 2

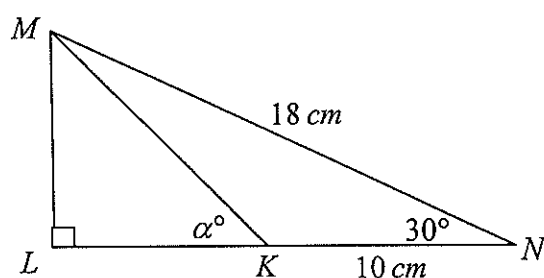
(ii) $y = \tan \theta$ 2

(B) Solve the following for $0^\circ \leq \theta \leq 360^\circ$:

(i) $\sqrt{2} \cos \theta = 1$ 2

(ii) $\tan(2\theta - 30^\circ) = 1$ 3

(C) Find α in the following diagram to the nearest degree. 3



END OF EXAMINATION

Yr 11 2010
2u Mem.
Half Yearly Solutions

Question 1

(a) 0.69

(b) $\left(\frac{1 + \sqrt{xy}}{x\sqrt{x}} \right) \times \frac{\sqrt{x}}{\sqrt{x}} = \frac{\sqrt{x} + x\sqrt{y}}{x^2}$

1 - multiplying
1 - correct ans.

(c) $2p^3 + 54 = 2(p^3 + 27)$
 $= 2(p+3)(p^2 - 3p + 9)$

— 1

— 1

(E) $|3x - 4| + 2 \leq 7$
 $|3x - 4| \leq 5$
 $-5 \leq 3x - 4 \leq 5$
 $-1 \leq 3x \leq 9$
 $-\frac{1}{3} \leq x \leq 3$

— 1 or equivalent

— 1 correct answer

(F) $a + b\sqrt{3} = \frac{\sqrt{3}}{3 + \sqrt{3}} \times \frac{(3 - \sqrt{3})}{(3 - \sqrt{3})}$ — 1
 $= \frac{3\sqrt{3} - 3}{9 - 3}$
 $= \frac{-3 + 3\sqrt{3}}{6}$
 $= \frac{-1 + \sqrt{3}}{2}$ — 1

$\therefore a = -\frac{1}{2} \quad b = \frac{1}{2}$ — 1

(D) $(h-2)^3 = h^3 + 3(h)^2(-2) + 3(h)(-2)^2 + (-2)^3$
 $= h^3 - 6h^2 + 12h - 8$

1 - correct terms

2 - correct terms
& signs.

* Please note: Marking Scheme
is a guideline only. Each student's
answers are considered on their
merits. Alternative responses may
also be acceptable.
2 - correct answer with rounding
1 - correct answer incorrect rounding

Solutions

Question 2

(A) (i) $D: x^2 - 4 \neq 0$

$(x-2)(x+2) \neq 0$

$\therefore x \neq \pm 2$ — 2
(1 each)

(ii) $D: 3 - x > 0$

$-x > -3$

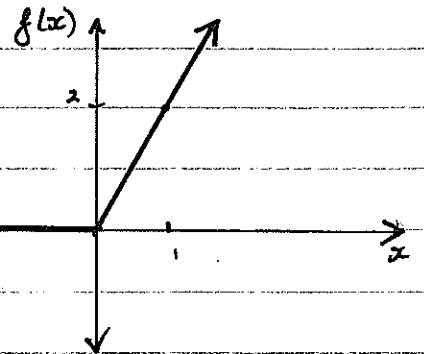
$x < 3$ — 1

(E) $|x| + x = x + x \quad x \geq 0$

$= 2x$

$= -x + x \quad x < 0$

$= 0$



1 for each branch.

(B) $g(x) = \frac{x^2}{x^4 + 1}$

(i) $g(-x) = \frac{(-x)^2}{(-x)^4 + 1}$

$= \frac{x^2}{x^4 + 1}$ — 1

(ii) $g(x) = g(-x)$ — 1

$\therefore g(x)$ is even

(C) $h(x) = x^2 + 3x$

(i) $h(-2) = (-2)^2 + 3(-2)$

$= -2$ — 1

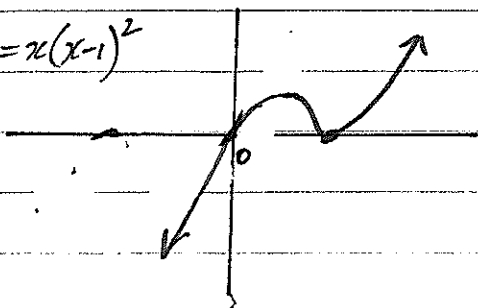
(ii) $h(a) = a^2 + 3a$ — 1

(iii) $h(x-1) = (x-1)^2 + 3(x-1)$ — 1

$= x^2 - 2x + 1 + 3x - 3$

$= x^2 + x - 2$ — 1

(D) $q(x) = x(x-1)^2$



or working

$\therefore q(x) \geq 0$ for $x \geq 0$ — 1

Question 4

(7) (i) $AC = \sqrt{(5-2)^2 + (-2-2)^2}$

$= \sqrt{49+16}$
 $= \sqrt{65}$ unit

(ii) $A(-3,2) \quad C(5,-2)$

$\frac{y-2}{x+2} = \frac{-2-2}{5-2}$

$\frac{y-2}{x+2} = \frac{-4}{3}$

$-4x-8 = 7y-14$

$4x+7y-6=0$ as required

or other correct method

(iii) $B(3,5)$ with $4x+7y-6=0$

$h \cdot d = \frac{|4 \times 3 + 7 \times 5 - 6|}{\sqrt{4^2 + 7^2}}$

$= \frac{41}{\sqrt{65}}$ or $\frac{41\sqrt{65}}{65}$

(iv) Area $\triangle ABC = \frac{1}{2} \times \sqrt{65} \times 41$

$= \frac{41}{2}$ unit² or other correct method
 $= 80\frac{1}{2}$ unit² — 1 method

(13) $3x-5y-3 + k(2x+3y+17)=0$

Sub (1,2) $3-10-3+k(2+6+17)=0$

$-10+25k=0$

$k = \frac{10}{25}$

$= \frac{2}{5}$

$\therefore 3x-5y-3 + \frac{2}{5}(2x+3y+17)=0$

$15x-25y-15 + 4x+6y+34=0$

$19x-19y+19=0$

$\therefore x-y+1=0$

(c) If $2x-3y-3=0$ is perp to $3x+ay+b=0$

Then $[a=2]$

$\therefore 3x+2y+b=0$

Now $x=3$ is the point of intersection of both lines

$\therefore 2 \times 3 - 3y - 3 = 0$

$6-3y-3=0$

$-3y=-3$

$y=1 \quad \therefore (3,1)$ must fit

$3 \times 3 + 2 \times 1 + b = 0$

$\therefore 11+b=0$

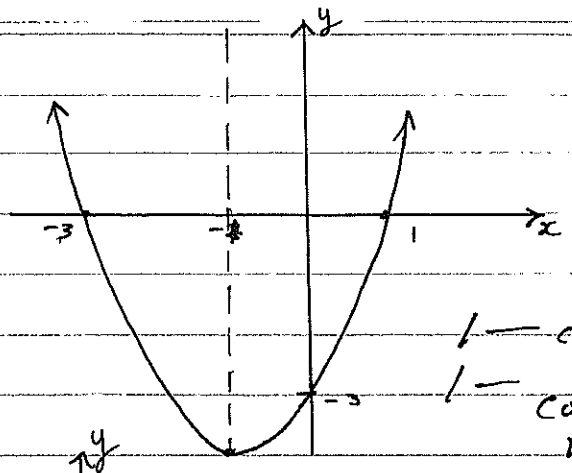
$b=-11$

$\therefore a=2, b=-11$

✓ ✓ ✓ ✓ ✓ ✓
1, 2, 3, 4, 5, 6

Question 3

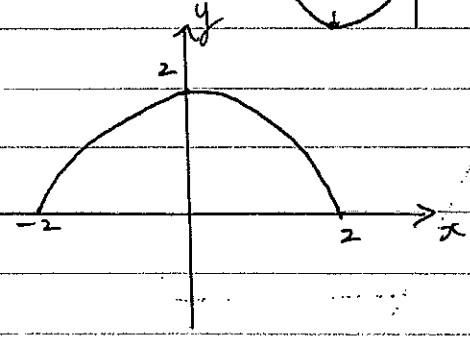
(A) (i) $y = x^2 + 2x - 3$
 $y = (x+3)(x-1)$



1 - correct shape
 1 - correct values

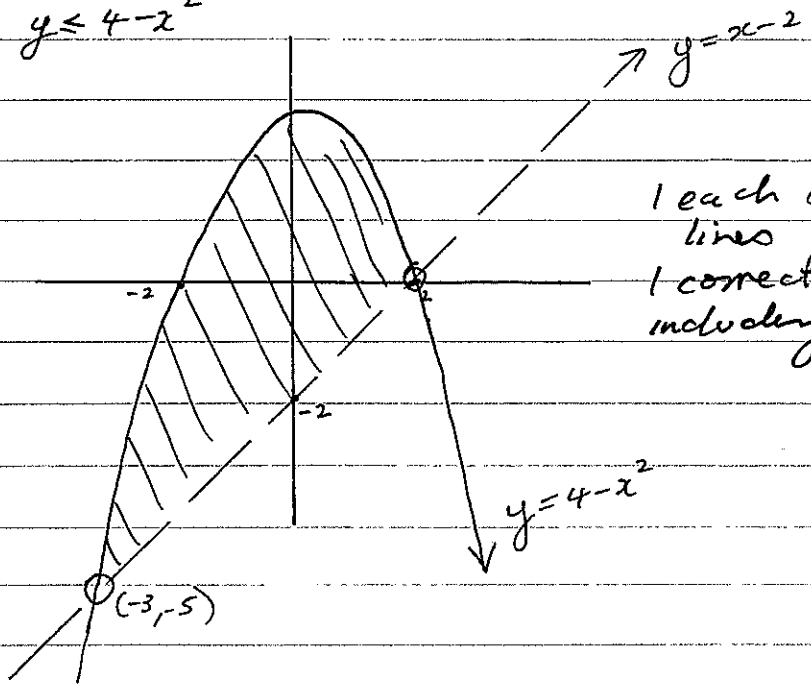
(ii) $y = \sqrt{4 - x^2}$

upper semi circle
 centre (0,0) radius 2 units



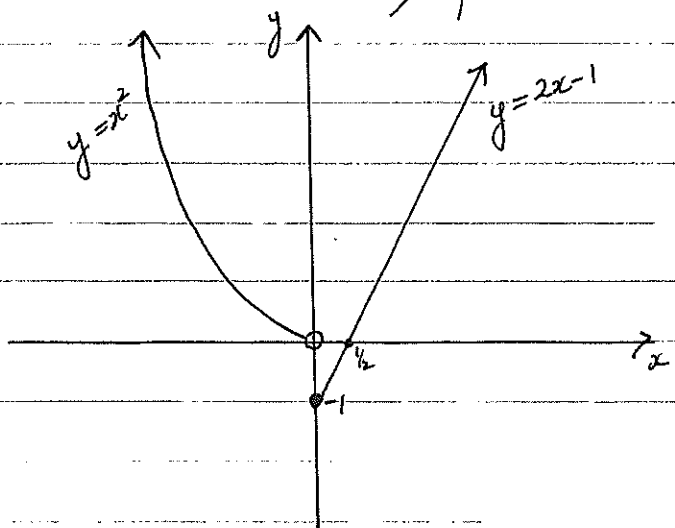
1 - correct shape
 1 - correct values

(B) $y > x - 2$ and $y \leq 4 - x^2$



1 each correct lines
 1 correct region including circles

(C)



$$f(x) = \begin{cases} 2x - 1 & \text{for } x \geq 0 \\ x^2 & \text{for } x < 0 \end{cases}$$

1 each correct lines
 1 correct end pts

Question 5

(A) (i) $\sin \theta = \frac{3}{4}$
 $\theta = 48^\circ 35'$ ——— |

(ii) $\sec \theta = 1.2230$
 $\cos \theta = \frac{1}{1.2230}$
 $\theta = 35^\circ 9'$ ——— |

disregard minute
rounding

(B) $\frac{135}{x} = \cos 16^\circ 24'$

$x = \frac{135}{\cos 16^\circ 24'}$ OR $135 \sec 16^\circ 24'$ ——— |

$= 140.7 \text{ cm (1 d.p.)}$ ——— |

(C) Let $\angle WAB = \theta$

$\therefore \cos \theta = \frac{175}{250}$

$\therefore$ Bearing of B from A
 is $270^\circ - \theta = 270^\circ - 46^\circ$
 $= 224^\circ$

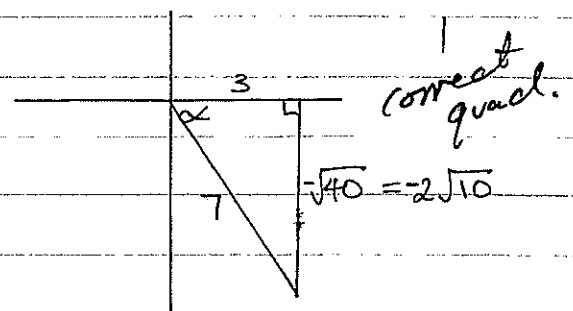
$\therefore \theta = 46^\circ$ (nearest deg) OR $S 46^\circ W$ ——— |

(D) (i) $\tan 210^\circ = -\tan 30^\circ$
 $= -\frac{1}{\sqrt{3}}$ ——— |

(ii) $\sin 330^\circ = -\sin 30^\circ$
 $= -\frac{1}{2}$ ——— |

(iii) $\sin 45^\circ \cos 120^\circ = \frac{1}{\sqrt{2}} \times -\frac{1}{2}$
 $= -\frac{1}{2\sqrt{2}}$ ——— |

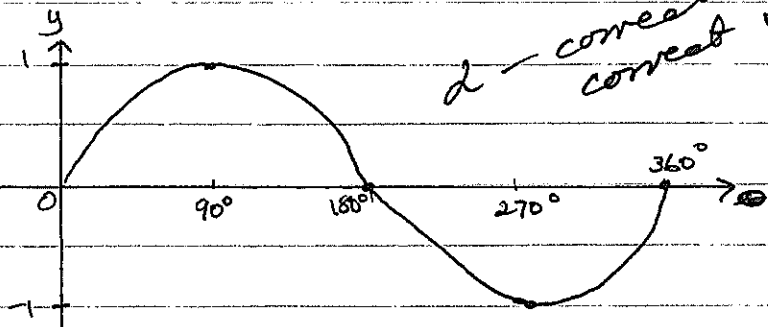
(E) $\sec \alpha = \frac{7}{3}$ $\sin \alpha < 0$
 $\therefore \cos \alpha = \frac{3}{7}$ $\therefore 4^{\text{th}} \text{ Quad}$



$\therefore \tan \alpha = \frac{-2\sqrt{10}}{3}$ ——— |

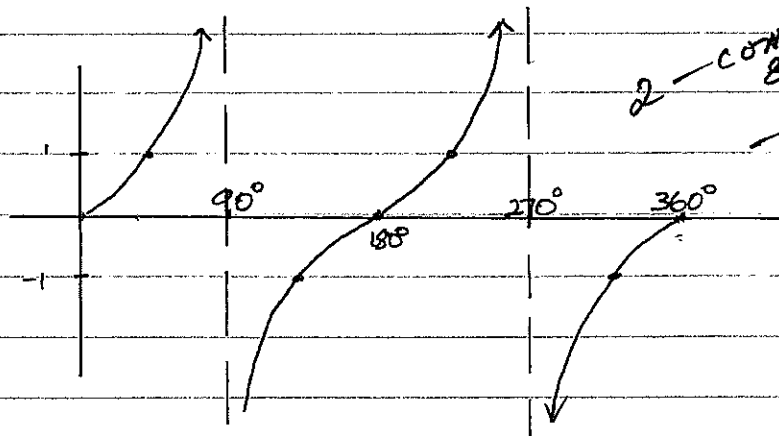
Question 6

(A) (i) $y = \sin \theta$



2 - correct shape
correct values

(ii) $y = \tan \theta$



2 - correct shape
correct values

(B) (i) $\sqrt{2} \cos \theta = 1$

$0^\circ \leq \theta \leq 360^\circ$

$\cos \theta = \frac{1}{\sqrt{2}}$

1st and 4th Quad.

$\theta = 45^\circ, 315^\circ$

(ii) $\tan (2\theta - 30^\circ) = 1$

$0^\circ \leq \theta \leq 360^\circ$

$2\theta - 30^\circ = 45^\circ, 225^\circ, 405^\circ, 495^\circ$

$0^\circ \leq 2\theta \leq 720^\circ$

$2\theta = 75^\circ, 255^\circ, 435^\circ, 525^\circ$

$-30^\circ \leq 2\theta - 30^\circ \leq 690^\circ$

$\theta = 37\frac{1}{2}^\circ, 127\frac{1}{2}^\circ, 217\frac{1}{2}^\circ, 262\frac{1}{2}^\circ$ — 3 for all 4 correct values

(C) $\frac{LN}{18} = \cos 30^\circ$

$\therefore LN = 18 \cos 30^\circ$

$= 18 \times \frac{\sqrt{3}}{2}$

$= 9\sqrt{3}$

So $\frac{LM}{LK} = \frac{9}{9\sqrt{3}-10} = \tan \alpha$

$\therefore \tan \alpha = \frac{9}{9\sqrt{3}-10}$

$\therefore \alpha = 58^\circ$ (nearest degree)

$\therefore \frac{LM}{9\sqrt{3}} = \tan 30^\circ$

$\therefore LM = 9\sqrt{3} \times \frac{1}{\sqrt{3}}$

$= 9$